

HOW TO WIRE THE AR-750

Read this sheet first. The next eight show you every wire.

AR-750 AGRICULTURAL ROVER

sheet 1 of 9

Four sheets of wiring, one map of the Pi's pins, two sheets of pictures so you can tell the parts apart, and a check list for the first time you switch it on. Nothing here was guessed: the pin numbers are read out of the rover's own settings file when this sheet is printed, and the sizes came out of the CAD model.

WIRE COLOURS

	RED	battery +, straight off the pack
	ORANGE	switched and fused 12 V
	GREEN	5 V from the buck converter
	BLACK	ground - all of it meets at one point
	YELLOW	a signal from the Pi
	BLUE	I2C, the two sensor wires

THREE THINGS THAT WILL BITE YOU

One common ground. The Pi's ground, both ESC grounds, the ADC, the receiver, the valve drivers and the battery minus all meet at ONE point. Most 'it moved on its own' faults are a missing ground.

The steering servo does not run off the Pi's 5 V pin. It pulls several amps against its stops and will reset the Pi. It goes on the buck.

A flyback diode across the pump and across every solenoid valve. Without them the collapsing coil resets the Pi, and you will spend an evening blaming the software.

AND THREE MORE

The ESCs are not H-bridges. They want a servo pulse, 1000 to 2000 us, and will not move until they have seen 1500 us for 2 seconds.

Both motors always get the same command. This is four wheel drive with a steering servo, not a tank. Never drive one axle faster to turn: the diffs let the inside wheels spin and it goes straight on.

Fuse the main lead and fuse each ESC. A 120 kg machine with a jammed wheel will happily set a wire on fire.

WHAT THE SHEETS ARE

2	Overview	the whole machine on one page
3	Power	battery to everything that draws current
4	Signals - driving	ESCs, steering, the spray boom
5	Signals - sensing	probe, sensors, camera, remote
6	The 40 pins	which hole each wire goes in
7-8	The parts	a picture of each one and what it does
9	First power on	the order to do it in, and what to check

EVERYTHING YOU NEED TO BUY

Sheets 7 and 8 show you what each one looks like.

x1	RASPBERRY PI 4	x5	LOGIC LEVEL MOSFET
x2	BTS7960 MOTOR DRIVER	x1	PI CAMERA
x2	DRIVE MOTOR	x1	RC RECEIVER
x1	STEERING SERVO	x1	MOTOR DRIVER, PROBE
x1	12 V 18 Ah LEAD ACID	x2	LIMIT SWITCH
x1	MAIN SWITCH	x1	SOIL MOISTURE SPIKE
x6	BLADE FUSES	x2	RESISTOR DIVIDER
x1	BUCK CONVERTER 5 V 5 A	x2	HEAD AND TAIL LIGHTS
x1	ADS1115 ADC	x1	ULTRASONIC RANGE SENSOR
x3	SOLENOID VALVE	x2	WHEEL ENCODERS
x1	DIAPHRAGM PUMP	x1	IMU

The last three are not fitted yet. The rover runs without them; sheet 9 says what each costs you.

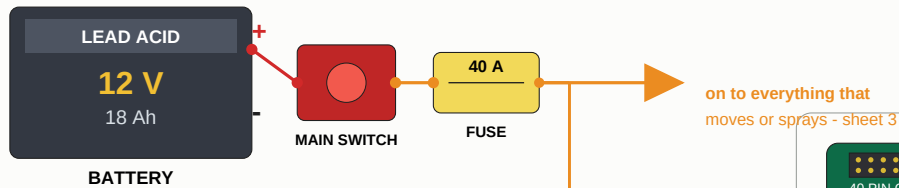
OVERVIEW - WHAT CONNECTS TO WHAT

AR-750 AGRICULTURAL ROVER

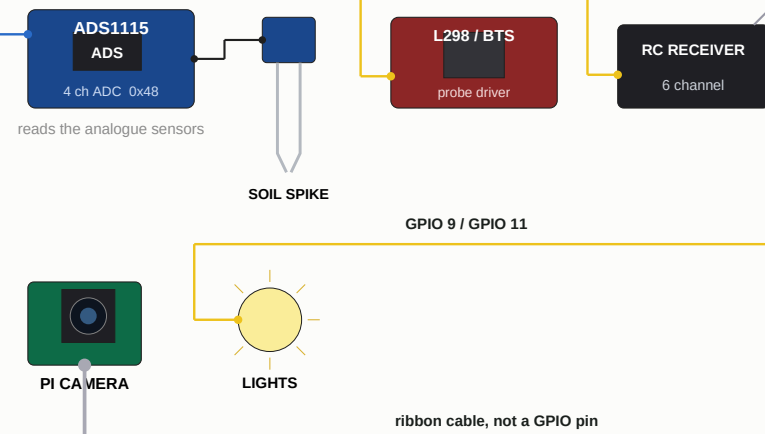
Block level. The next sheets give you the actual wires.

sheet 2 of 9

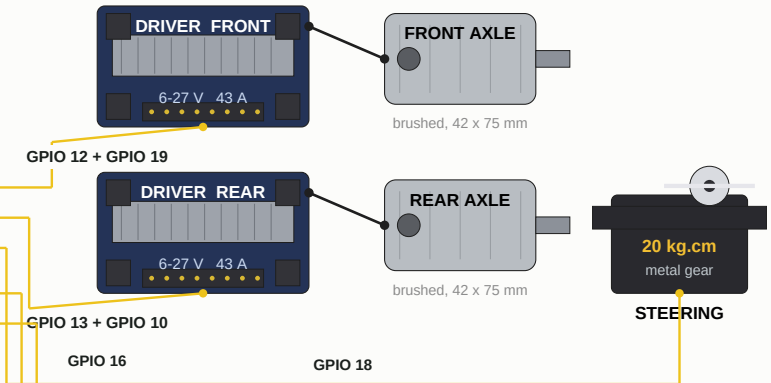
POWER



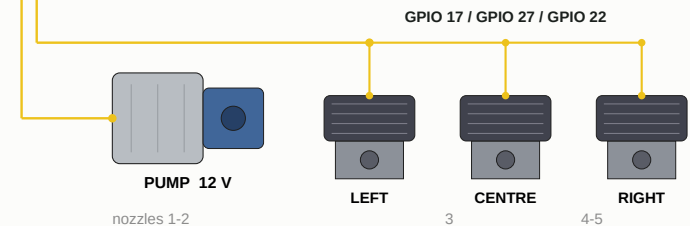
SENSING AND CONTROL



DRIVE - four wheel drive



SPRAY BOOM - three sections, 5 nozzles



TWO THINGS ON THIS PAGE

The camera is the only thing that does not use a GPIO pin - it plugs into the flat white ribbon socket next to the 5 V connector.

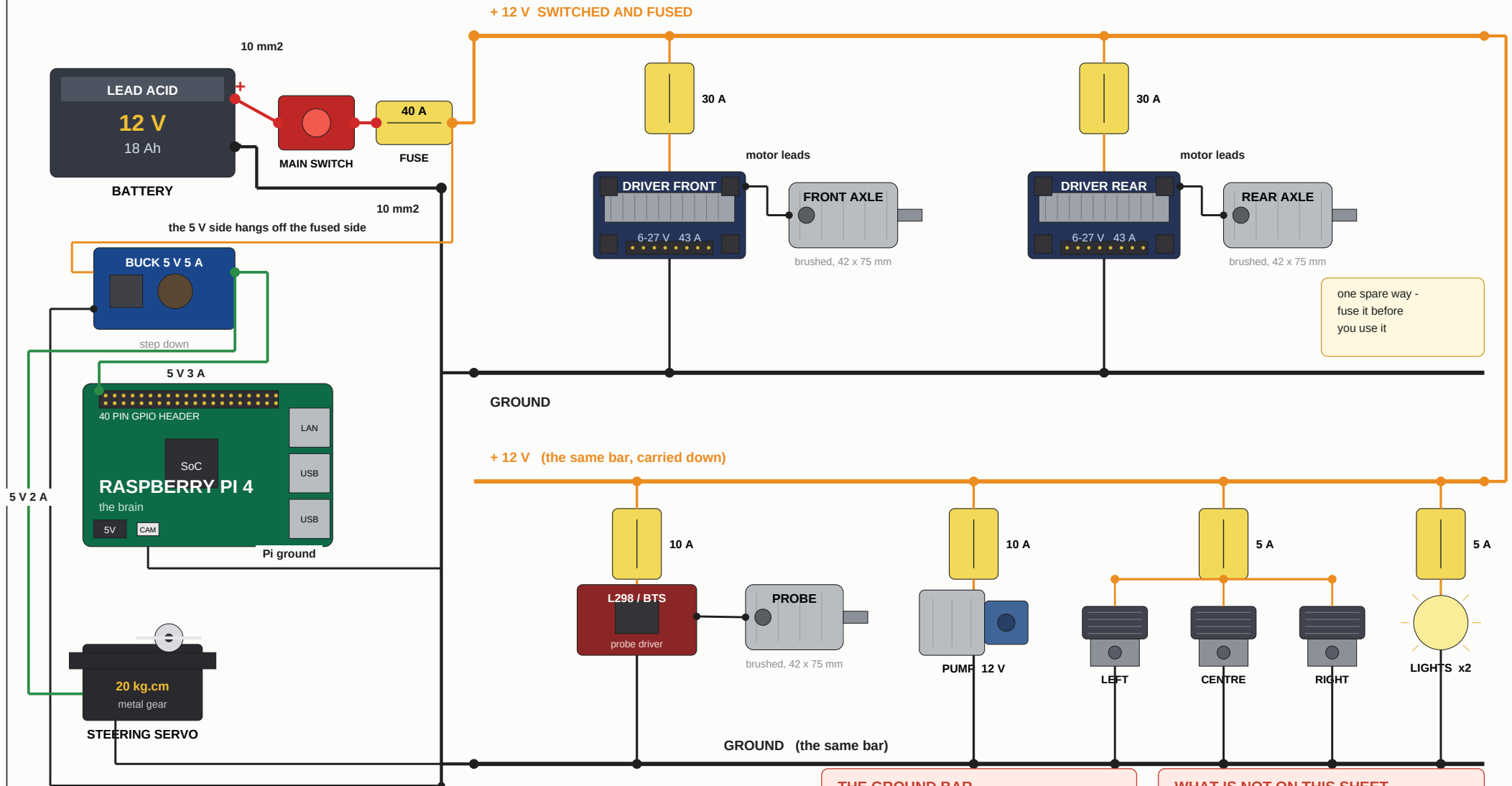
The remote always wins. Touch a stick while it is driving itself and it drops into manual on the spot, mid row, mid spray.

POWER

AR-750 AGRICULTURAL ROVER

sheet 3 of 9

Fat wire. Everything on this sheet can start a fire if you get it wrong.



The servo takes its 5 V from the buck, NOT from the Pi's 5 V pin. It pulls 3 A against its stops and resets the Pi.

WIRE SIZES

Main lead and both driver leads: 10 mm² silicone.
Pump and probe: 2.5 mm².
Valves and lights: 1 mm².

THE GROUND BAR

ONE piece of brass or copper. Never daisy chain grounds part to part - one loose screw then takes out everything after it.

WHAT IS NOT ON THIS SHEET

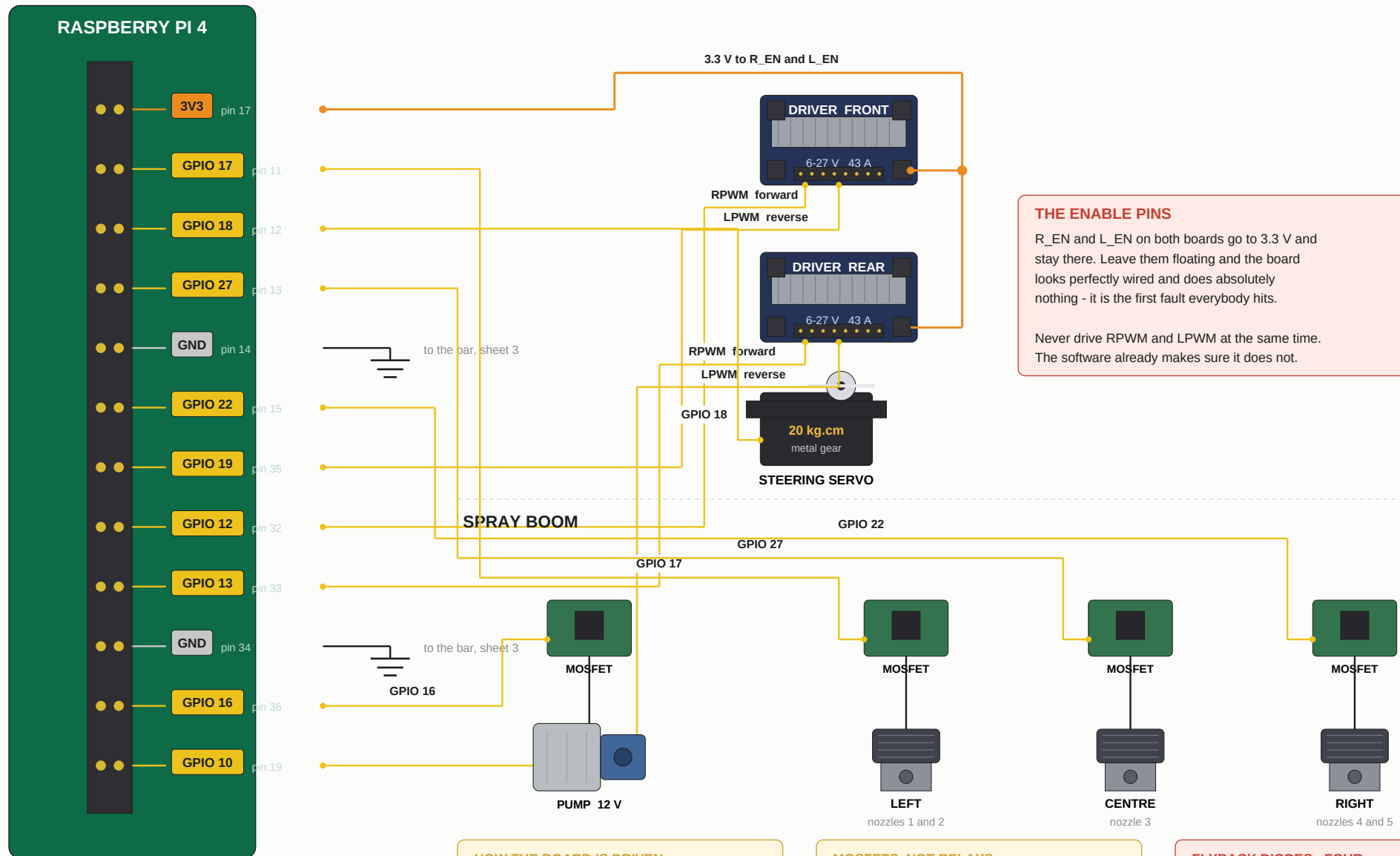
The MOSFET that switches the pump, each valve and each light sits in the ground leg - see sheet 4. A flyback diode across each coil.

SIGNALS - DRIVING AND SPRAYING

AR-750 AGRICULTURAL ROVER

Thin wire. Signal and ground only - nothing on this sheet carries any real current.

sheet 4 of 9



THE ENABLE PINS

R_EN and L_EN on both boards go to 3.3 V and stay there. Leave them floating and the board looks perfectly wired and does absolutely nothing - it is the first fault everybody hits.

Never drive RPWM and LPWM at the same time. The software already makes sure it does not.

HOW THE BOARD IS DRIVEN

Two pins per motor, both PWM at 1000 Hz. Forward puts PWM on RPWM and holds LPWM low. Reverse swaps them. Both low is coast.

MOSFETS, NOT RELAYS

The valves pulse about 10 times a second and the software sets how long each stays open. A relay cannot switch that fast.

FLYBACK DIODES - FOUR

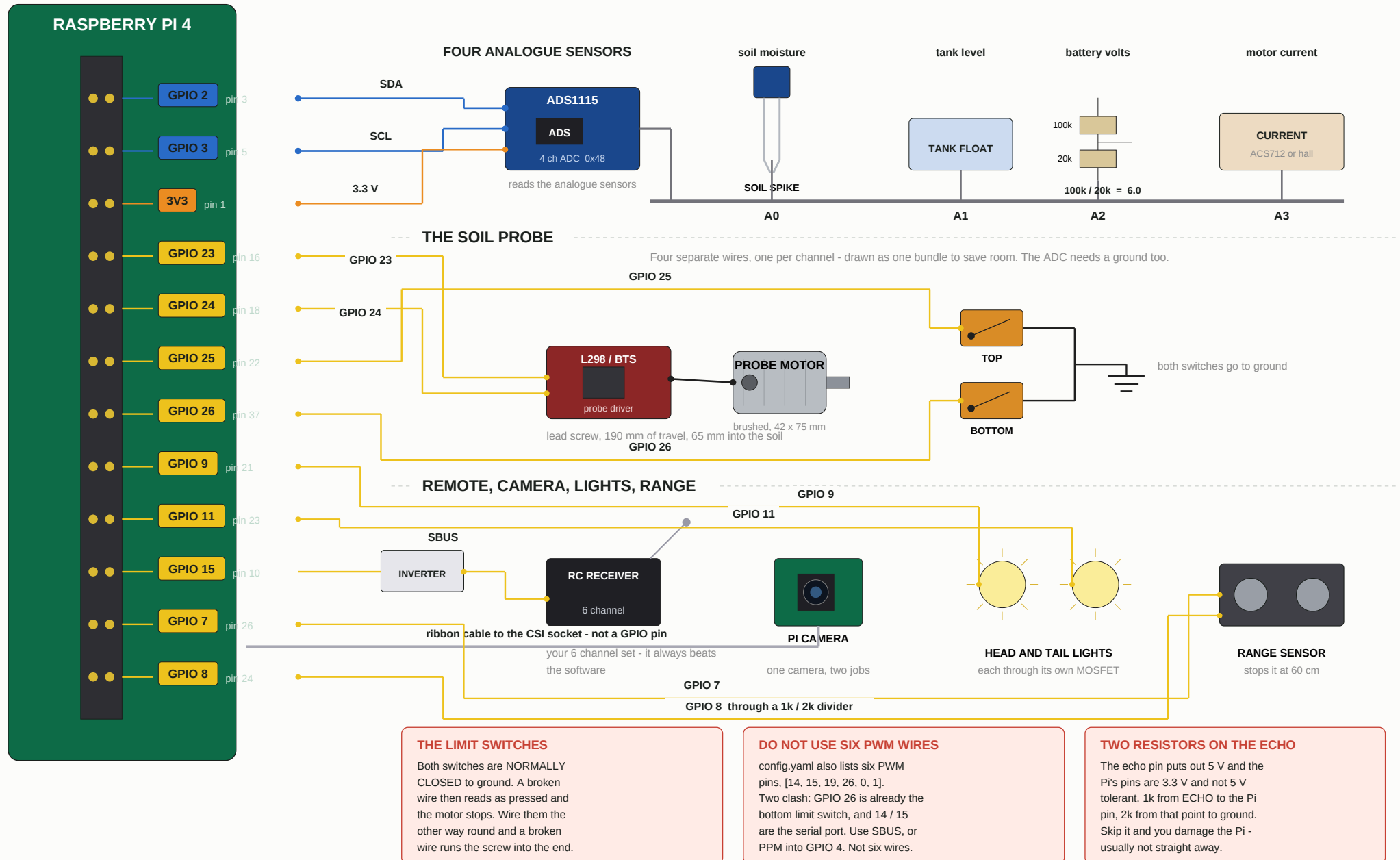
A 1N5408 across each coil, band to the + side. The pump and all three valves - four diodes. The difference between working and random resets.

SIGNALS - SENSING, REMOTE, LIGHTS

AR-750 AGRICULTURAL ROVER

The probe, the sensor board, the camera, and your 6 channel set.

sheet 5 of 9



THE 40 PINS ON THE RASPBERRY PI

AR-750 AGRICULTURAL ROVER

Which hole each wire goes in. Pin 1 is the one nearest the SD card corner.

sheet 6 of 9



COLOUR KEY

- USED** the AR-750 needs this wire
- I2C** the two sensor wires
- POWER** 3.3 V and 5 V out of the Pi
- GROUND** use several, not just one
- LATER** wheel encoders when you fit them
- FREE** nothing of ours on it

FOUR DRIVE PINS, NOT TWO

The two driver boards take FOUR pins, not two: one for forward and one for reverse on each motor. GPIO 12 and GPIO 19 are the front board, GPIO 13 and GPIO 10 the rear.

GPIO 18 stays the steering servo. It is one of only three pins on a Pi that can make a rock steady servo pulse, so leave it there.

SPREAD THE GROUNDS OUT

Use more than one ground pin. Pins 6, 9, 14, 20, 25, 30, 34 and 39 are all ground and they are spread along the header on purpose.

Put the drivers' ground on pin 34, sensors on pin 9, the boom on pin 14. Then one loose wire is one system, not the whole rover.

Run `pinout` on the Pi at any time and it prints this same picture in the terminal.

Pin numbers are BCM (the GPIO number). Every one of them comes from `ar750/config.yaml` - if you move a wire, change that file, not the code.

WHAT EVERY PART LOOKS LIKE (1 of 2)

So you can tell them apart in the box. The number in the corner is how many you need.

AR-750 AGRICULTURAL ROVER

sheet 7 of 9

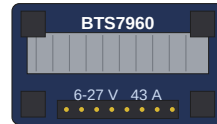


x1

RASPBERRY PI 4

The brain. Runs the rover software and serves the website you open on your phone.

4 GB is plenty. It needs a solid 5 V 3 A supply.

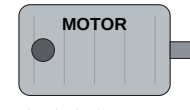


x2

BTS7960 MOTOR DRIVER

Turns the Pi's two PWM pins into motor power, both directions. One per axle.

6-27 V, 43 A. Tie R_EN and L_EN to 3.3 V or nothing moves at all.



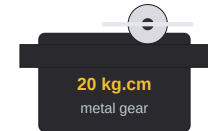
brushed, 42 x 75 mm

x2

DRIVE MOTOR

One drives the front differential, one the rear. Always the same command.

42 x 75 mm in the model, through the 5.94:1 gear train.

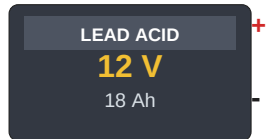


x1

STEERING SERVO

Points the front wheels. Full travel gives 24.7 deg at the wheel.

20 kg.cm metal gear minimum. Set its stops BEFORE you fit the drag link.



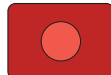
x1

BATTERY

12 V 18 Ah LEAD ACID

Everything on the rover comes from this one pack. Cheapest sensible battery by a mile.

The software stops the motors at 11.0 V to protect it. Set `safety.battery.chemistry` to `match whatever you fit`.



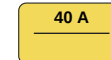
x1

MAIN SWITCH

MAIN SWITCH

Kills everything. Mount it where you can reach it while walking beside the rover.

Rated 60 A or more. A little toggle switch will weld shut.



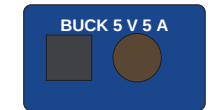
x6

FUSE

BLADE FUSES

40 A on the main and on each ESC, 10 A on the pump and the probe, 5 A on valves and lights.

In holders you can reach without taking the lid off.



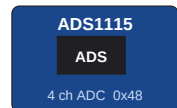
x1

step down

BUCK CONVERTER 5 V 5 A

Drops 12 V to 5 V for the Pi and the steering servo.

5 A, not 3. The servo alone can pull 3 A against the stops.



x1

reads the analogue sensors

ADS1115 ADC

The Pi cannot read an analogue sensor. This board can, on four channels.

I2C address 0x48. Soil, tank, battery, motor current.



x3

SOLENOID VALVE

One per boom section. They pulse about ten times a second to set the rate.

12 V, normally closed. A flyback diode across each one.



x1

DIAPHRAGM PUMP

Pressurises the spray line. Cannot run until you arm it on the remote or the website.

12 V, about 4 l/min. Flyback diode across it.



x5

MOSFET

LOGIC LEVEL MOSFET

The Pi's pin cannot switch a pump or a valve. One of these does it for each.

IRLZ44N or similar - it must switch fully on at 3.3 V.

WHAT EVERY PART LOOKS LIKE (2 of 2)

AR-750 AGRICULTURAL ROVER

So you can tell them apart in the box. The number in the corner is how many you need.

sheet 8 of 9



PI CAMERA

x1

PI CAMERA

Follows the row AND judges the plants.
Lens sits 254 mm up, 95 mm ahead of the front axle.

Ribbon cable to the CSI socket, not a GPIO pin. Tilt it down 12 deg.



RC RECEIVER

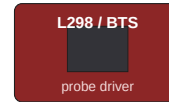
6 channel

x1

RC RECEIVER

Your existing 6 channel set. It always beats the software.

SBUS through an inverter into GPIO 15, or PPM into GPIO 4.



L298 / BTS

probe driver

x1

MOTOR DRIVER, PROBE

A small H-bridge so the probe's lead screw can run both ways.

L298N or a BTS7960. 5 A is plenty for this job.



x2

LIMIT SWITCH

One at each end of the probe's travel. They stop the motor.

Wired NORMALLY CLOSED to ground, so a broken wire stops it too.



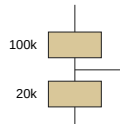
SOIL SPIKE

x1

SOIL MOISTURE SPIKE

Goes 65 mm into the soil on the lead screw and reads the moisture.

Capacitive, not the cheap resistive kind - those corrode in weeks.



100k

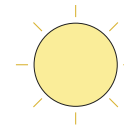
20k

x2

RESISTOR DIVIDER

One lets the ADC see the 16.8 V battery. One protects the Pi from the range sensor's 5 V echo.

Battery 100k / 20k. Echo 1k / 2k.



x2

HEAD AND TAIL LIGHTS

So you can see what it is doing at dusk, and so the camera has light.

12 V LED bars, each through its own MOSFET.



HC-SR04

x1

ULTRASONIC RANGE SENSOR

Stops the rover before it drives into a person or a wall.

Not switched on in the config yet - turn it on once it is wired.



slotted disc

x2

WHEEL ENCODERS

Not fitted. Without them the distance on the dashboard is worked out from what the motors were told, and it drifts.

The cheapest worthwhile upgrade on the machine. The code already supports them.



MPU-6050

x1

IMU

Not fitted. Without it the tilt cut-out can never fire, and this machine is 404 mm tall on a 430 mm track.

MPU-6050, on the same two I2C wires, address 0x68.

1 Wire the power only

Battery, switch, fuse, ground bar, buck converter. Nothing else.
Put a meter on the buck output and set it to 5.1 V before anything is plugged into it.

2 Add the Pi, on its own

5 V from the buck to pins 2 and 6. Boot it, copy the software over,
run `bash install.sh`, then `python3 -m ar750.main --sim`.
Open the website it prints. Sign in admin / agrirover and change it.

3 ESCs and motors, ON BLOCKS

All four wheels off the ground. Red BEC wire out of both ESC plugs.
python3 scripts/bench.py wheels
If an axle runs backwards set `invert_front` or `invert_rear` in the config file. Do not swap the motor wires.

4 Steering, drag link OFF

Servo powered from the buck, signal to GPIO 18.
python3 scripts/bench.py steering
Check the horn does not hit its stops anywhere in the sweep, THEN connect the drag link. A servo fighting its stops strips in seconds.

5 The remote

python3 scripts/bench.py rc
Move each stick and watch which channel moves. Fix the map in the config file if yours is wired differently. Check the failsafe by switching the transmitter off - it must stop.

6 The probe, nothing underneath

python3 scripts/bench.py probe
Press each limit switch by hand first and watch the software see it.
Both legs should take about the same time.

7 The boom, WITH WATER

python3 scripts/bench.py boom
It tells you what nozzle size you actually need. Read that before you buy nozzles - see the warning on the right.

8 Sensors

python3 scripts/bench.py adc
Compare the battery reading against a meter. If it is out, your divider is not exactly 6.0 - put the real number in the config.

READ THIS BEFORE YOU BUY NOZZLES

0.6 x (ml/min per nozzle)
litres/ha = -----
km/h x spacing in metres

This rover follows a row at 0.16 m/s (0.57 km/h).
At 140 mm spacing, the 350 ml/min nozzles in the config would put down about 2646 l/ha wide open. A vegetable crop wants around 200. That is 13 times too much, and you cannot drive fast enough down a bed to fix it.

Buy the smallest nozzles you can get - about 26 ml/min - and let the valves pulse. With the ones in the config they would sit at only 8% duty, which goes stripey.

bench.py boom does this sum for whatever you fit.

WHAT IT CANNOT DO, SO YOU ARE NOT SURPRISED

It cannot turn on the spot. The turning circle is 2.78 m, so leave yourself a headland at the end of every bed.

There is no trained AI model in the software yet. Until you train one it photographs every plant and logs it as 'not checked'. It will not guess a disease it did not see.

There is no arm on this machine. A weed gets photographed and logged for you, not pulled.

The red stop button on the website goes through software. A real stop button, wired to cut both ESC signal lines, does not care whether the software is still alive. Fit one.